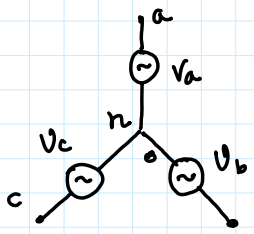
Lecture 23: Three-phase Circuits

- **Three-phase** circuits are common in electrical industry, all over world. The generation, transmission of electrical energy is most efficiently done at higher voltage 3ϕ systems.
- Three-phase generation, transmission is widely accepted from the economics and design considerations of transmission lines, electrical machines and transformers etc.

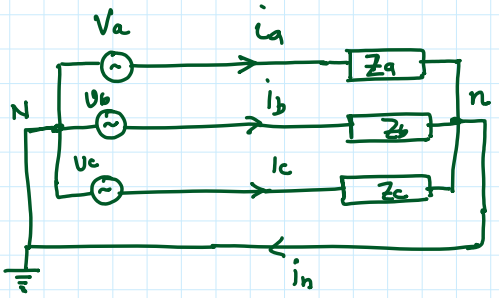
Comparison of Single-phase and Three-Phase System

Aspects	Single-phase Circuits	Three-phase AC Circuits
Wire configuration	The single-phase supply requires two wires (one phase and one neutral) for completing the circuit.	The three-phase requires three phase wires and one neutral wire for completing the circuit.
Power transfer	Power transfer is less efficient due to oscillating component.	Three times more power transfer at high efficiency due balanced voltages and currents
Supply during Fault conditions	The single-phase system has only one phase wire, and if the fault occurs on the network, then the power supply completely fails.	In three-phase system the network has three phases, and if the fault occurs on any one of the phases, the other two will continuously supply the power.
Efficiency	The efficiency of the single-phase supply is less as compared to three phase supply. For 2 wire, the power supply is P=VI cos ϕ.	Three- phase supply requires less conductor as compared to single phase supply for the equivalent power delivery For 4 wires supply, the power supply is 3 P = 3VI cos ϕ Thus, the material efficiency is 1.5 times more.
Maintenance	The single-phase supply requires more maintenance and become costly.	Three-phase supply needs less maintenance and is more rugged system
Applications	The single-phase supply is mostly used in the house and for running the small loads.	The three-phase supply is used in large industries and for running the heavy loads.
Usage	The single-phase loads constitute 10% lighting,	The three-phase load forms about 70-75% due to 3-phase induction motors, of total load, 10-15% for industrial heating, furnace heating and cooling.

Balanced 3-ϕ system: A 3ϕ balanced system is shown in figure below.



Three-phase balanced system



$Z_a = Z_b = Z_c = Z$

If the neutral is grounded ⇒ 3ϕ 4 wire system
not grounded ⇒ 3ϕ 3 wire system

Representing Three-phase System in Time Domain and Phasor Form

Time domain

$$v_a(t) = \sqrt{2}V \sin \omega t$$
$$v_b(t) = \sqrt{2}V \sin(\omega t - 120^\circ)$$
$$v_c(t) = \sqrt{2}V \sin(\omega t + 120^\circ)$$

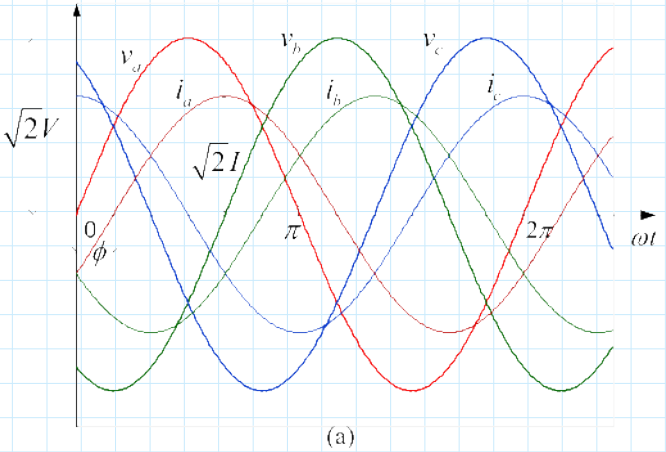
$$i_a(t) = \sqrt{2}I \sin(\omega t - \phi)$$
$$i_b(t) = \sqrt{2}I \sin(\omega t - 120^\circ - \phi)$$
$$i_c(t) = \sqrt{2}I \sin(\omega t + 120^\circ - \phi)$$

Phasor:

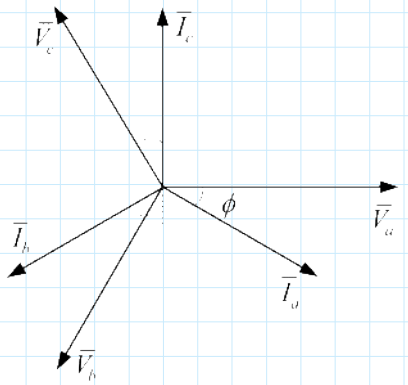
$$\bar{V}_a = V \angle 0$$
$$\bar{V}_b = V \angle -120^\circ$$
$$\bar{V}_c = V \angle 120^\circ$$

$$\bar{I}_a = I \angle -\phi$$
$$\bar{I}_b = I \angle -120^\circ - \phi$$
$$\bar{I}_c = I \angle 120^\circ - \phi$$

a-b-c
RYB – Red – Yellow – Blue, Black – Neutral, Green: Earth
TUV



(a)



(b)

$$p_{3\phi}(t) = v_a i_a + v_b i_b + v_c i_c$$

$$= VI\left[2\sin\omega t \sin(\omega t - \phi) + 2\sin(\omega t - 120^\circ)\sin(\omega t - 120^\circ - \phi) + 2\sin(\omega t + 120^\circ)\sin(\omega t + 120^\circ - \phi)\right]$$

$$= VI\left[\cos\phi - \cos(2\omega t - \phi) + \cos\phi - \cos\{2(\omega t - 120^\circ) - \phi\} + \cos\phi - \cos\{2(\omega t + 120^\circ) - \phi\}\right]$$

$$= VI\left[3\cos\phi - \{\cos(2\omega t - \phi) + \cos(2\omega t - \phi + 120^\circ) + \cos(2\omega t - \phi - 120^\circ)\}\right]$$

$\cos\theta + \cos(\theta + 120^\circ) + \cos(\theta - 120^\circ) = 0$

$$p_{3\phi}(t) = 3VI\cos\phi = P_{3\phi}$$

$$= \frac{1}{T}\int_0^T (v_a i_a + v_b i_b + v_c i_c)\, dt$$

$$= \frac{1}{T}\int_0^T (V_a i_a + V_b i_b + V_c i_c)\, dt$$

Thus real power $P = 3VI\cos\phi = p_{3\phi}(t)$

$$Q \triangleq 3VI\sin\phi$$

$$S = 3VI,$$

$$pf = \frac{P}{S} = \cos\phi$$

Complex Apparent Power: $\bar{S} = \bar{V}_a \bar{I}_a^* + \bar{V}_b \bar{I}_b^* + \bar{V}_c \bar{I}_c^*$

$$= V\angle 0 \; I\angle -\phi + V\angle -120^\circ \cdot (I\angle -120^\circ - \phi)^* + V\angle 120^\circ (I\angle 120^\circ - \phi)^*$$

$$= VI\angle\phi + VI\angle\phi + VI\angle\phi = 3VI\cos\phi + j\,3VI\sin\phi$$

$$= P + jQ$$

$$\Rightarrow P_{3\phi} = 3VI\cos\phi = p_{3\phi}(t)$$

and $Q_{3\phi} = 3VI\sin\phi$

This is quite interesting fact that in 3ϕ balanced system, the 3ϕ instantaneous power is constant. The time varying components get cancelled. With respect to single phase system, where we have $p_{1\phi}(t) = VI\cos\phi - VI\cos(2\omega t - \phi)$ ---- (#)

$$p_{3\phi}(t) = 3V\cos\phi \quad \text{---------- (\#\#)}$$

There is no oscillating component of power that move to and foo between the source and load. This improves the following points.

Starting torque in 3ϕ induction machine
Smooth operation, ripple free torque
High power factor and efficiency.

Smooth operation, ripple free torque
High power factor and efficiency.

Due to this: single-phase motors are practically limited to fractional horse power to few kilowatts. Such as household applications, refrigerator, water pumps, air-conditioners etc.

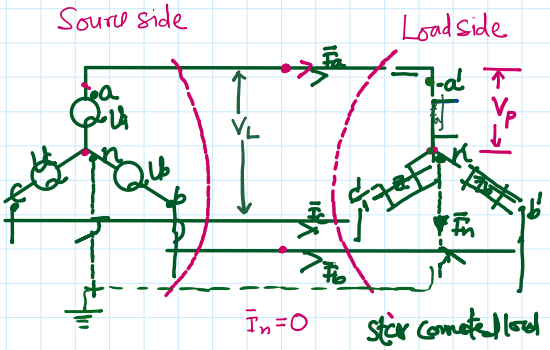
3ϕ machines/ motors are used for heavy industrial works starting from kilowatts to megawatt ratings, in steel plants, semiconductor industries and many others.

Line and Phase Quantities

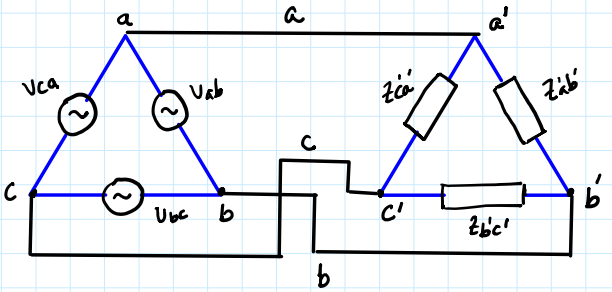
The load in three-phase system may be connected in star or delta as shown below. Accordingly we have line and phase quantities as shown in figure below.

Y - Y
Y - Δ
Δ - Y
Δ - Δ

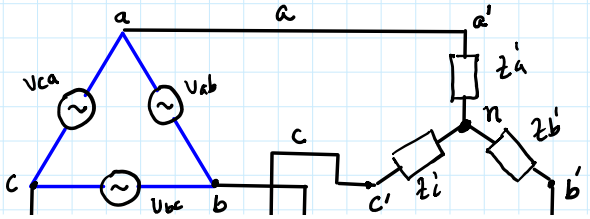
Each configuration has phase and line, voltage and current quantities. And there is relationship between the phase and line quantities of each configuration.

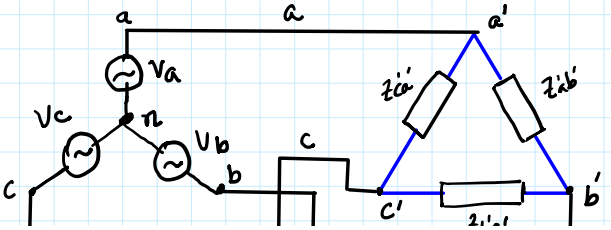


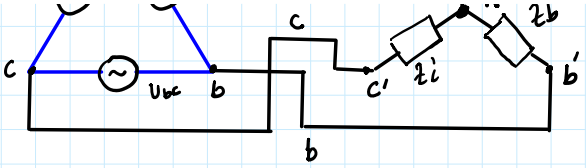
Star-star Configuration.



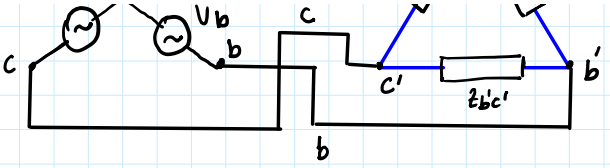
Delta-delta Configuration







Delta - Star configuration



Star - delta configuration